

Space

'Impossible' star mystery resolved by three-body solution

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THE conundrum of a star that seemed too small to exist has been solved by the detection of another star hiding in the same system.

In 2019, astronomers announced the discovery of a system called KIC 8145411 in which a white dwarf – the dull, compact remnant left behind after certain stars exhaust their nuclear fuel – was orbiting a sun-like star every 450 days. The system is a rare example of a self-lensing binary, where the gravitational heft of the compact white dwarf magnifies the light of its companion star relative to us, allowing astronomers to measure the dwarf's mass.

They found that the white dwarf was just a fifth the mass of our sun, which is extremely low for such a star, and was previously only seen in situations where the dwarf tightly orbits another star that saps mass from it. In KIC 8145411, the white dwarf orbits at a distance slightly greater than that between Earth and our sun (1 astronomical unit, or AU), which is too far for the larger star to chip away at the dwarf. Astronomers concluded that this impossible star shouldn't exist.

Now, Natsuko Yamaguchi at the California Institute of Technology and her colleagues have taken another look, using the Palomar Observatory in California. They found that the system contains another sun-like star, orbiting the other two at about 700 AU away. Factoring this in, the team calculated the mass of the white dwarf as about half that of our sun (arXiv, doi.org/mv39). "It's no longer impossible," says Yamaguchi.

The hidden star discovery hints that other supposed binary systems may host extra stars, confusing some calculations for working out stellar masses. "This situation where you have two bright stars but you think it's one is probably not that uncommon," says Yamaguchi. ■

Analysis Mental health

Does using the internet make us happier or sadder? A large study suggests that access to the internet is linked to our life satisfaction, but questions remain, says Clare Wilson

GROWING access to the internet is linked with better well-being around the world, a large study has concluded. The finding seems to be at odds with concerns that smartphones have led to a flood of mental health problems. So, where does the truth lie?

Concerns over excessive use of phones may seem a modern phenomenon, but they echo similar fears over past tech trends, such as that people would get hooked on computer games or TV, says Andrew Przybylski at the University of Oxford.

Yet the widespread adoption of smartphones does seem a more extreme social phenomenon. It has also coincided with rising levels of mental health problems in younger people in recent years, especially teenage girls.

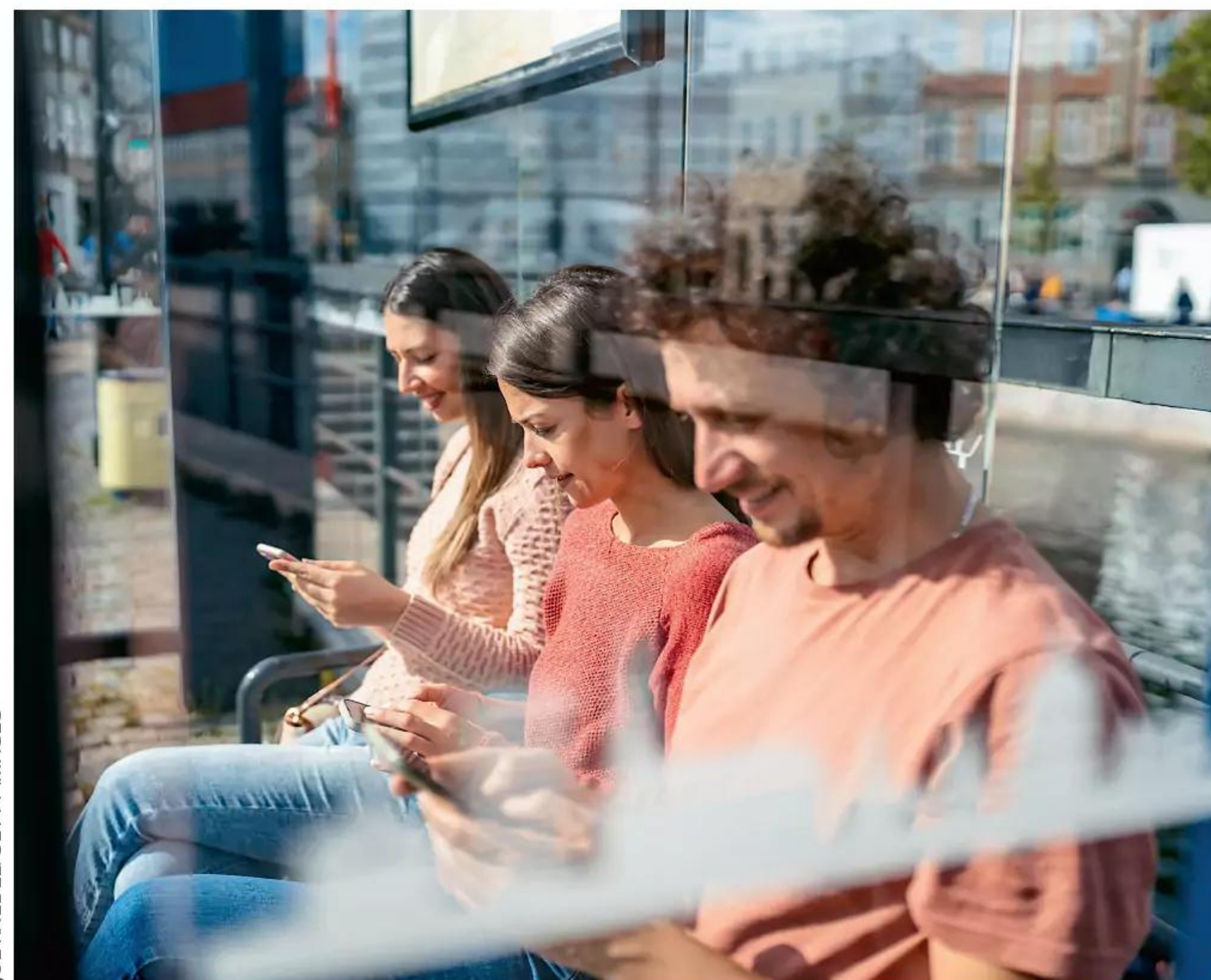
Most studies that have tried to investigate whether the two trends are linked have looked at people in Western countries. Instead, Przybylski and his colleague Matti Vuorre at Tilburg University in the Netherlands used data from the long-running Gallop World Poll, which included questions about the use of the internet and internet-connected phones.

2m+

Number of people polled between 2006 and 2021

The survey asked people about eight indicators of well-being, including life satisfaction and positive or negative experiences, as well as collecting demographic data. The researchers analysed figures for more than 2 million people from 168 countries between 2006 and 2021.

The pair carried out more than 30,000 analyses for each country and year. The vast majority found that people were more likely to



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What we see on our smartphones influences our lives

have higher well-being if they had internet access. These positive associations remained even when the analyses were adjusted to account for people being more likely to use the internet if they had a higher income or level of education (*Technology, Mind, and Behavior*, doi.org/mv6v).

While this kind of analysis couldn't prove that internet access increases people's happiness, it may help them make friends or have access to more job opportunities, says Przybylski. "These are social technologies."

The findings are unlikely to settle any arguments, though. For one thing, the primary concern about the harms of social media relates to its impact on children, teenagers and young adults, such as whether it may facilitate bullying, promote unrealistic body images or allow radicalisation.

The Gallop poll covered people aged between 15 and 99, so it is possible that any harms specific

to younger people were masked by the fact that older people could be more resilient.

Another issue is that the survey questions about internet use required a simple yes or no answer – they didn't ask people how long they spent using it. It is possible that some access to the internet is better for your mental health than none, but that excessive use is harmful, says Andres Roman-Urrestarazu at the University of Cambridge. "I'm not sure the type of measurements they're using are adequate."

So, the new findings may tell us that a farmer in Kenya becomes happier when they can use their phone to check which markets will give them the best price for their produce, but it doesn't tell parents in the UK if they should let their pre-teen start using WhatsApp.

Nevertheless, the findings bring a new slant to the debate about how the internet is changing the world and is a reminder that new media can be beneficial in many countries. "It really does open you up to a lot of good stuff, [as well as] scams and different types of exploitation," says Przybylski. ■